

Exercise 1

Let $\star$ be an internal law defined on $\mathbb{R}$ by :

$$\forall (x, y) \in \mathbb{R}^2 \quad x \star y = x + y + x^2 y^2$$

- 1) Verify that $\star$ is commutative.
- 2) Is $\star$ associative ?
- 3) Show that $\mathbb{R}$ admits a neutral element for $\star$.
- 4) Solve the following equations in $\mathbb{R}$

$$1 \star x = 0, \quad 1 \star x = 1$$

Exercise 2

Let $G = \mathbb{R}^* \times \mathbb{R}$ and let $\star$ be an internal law in G defined by

$$(x, y) \star (x', y') = (xx', xy' + y)$$

- 1) Show that $(G, \star)$ is a group, is it commutative ?
- 2) Show that $(]0, +\infty[\times \mathbb{R}, \star)$ is a subgroup of $(G, \star)$.

Exercise 3

Determine which of the sets E_1, E_2, E_3 are vector subspaces of $\mathbb{R}^3$

$$E_1 = \{(x, y, z) \in \mathbb{R}^3, 3x - 5y = 0\}$$

$$E_2 = \{(x, y, z) \in \mathbb{R}^3, x^2 - y^2 = 0\}$$

$$E_3 = \{(x, y, z) \in \mathbb{R}^3, x + y = 0 \text{ and } z = 0\}$$

Exercise 4

We consider two vector subspaces E, G of $\mathbb{R}^3$ defined by :

$$E = \{(x, y, z) \in \mathbb{R}^3, x = 0, y = z\}$$

$$G = \{(x, y, z) \in \mathbb{R}^3, y = 0\}.$$

Give its dimensions.

Exercise 5

We consider an application $f : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ defined by :

$$f(x, y, z) = (x + y, x - z, y - z)$$

- 1) Show that f is linear.
- 2) Determine $\text{Ker } f$ (kernel of f) and $\text{Im } f$ (image of f).